Learning from Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Learning from Examples before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Learning from Examples trace and record the first step you can explain without help.

Use W2: Explain a tied fit for **Fitting** a **pattern** from examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Learning from Examples explanation against the supplied facts before discussing it.

Use the worked case for **Fitting** a **pattern** from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Fit a new distance task

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

> **My independent explanation, trace and limits**
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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Learning from Examples target to a trace and a justified conclusion.

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Learning from Examples case and show the deciding evidence.

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

Predict and state the **evidence** limit.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A learner sees three red apples and no green apples, then calls a green apple not an apple. What missing **evidence** could help? How would you check improvement?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 13 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Learning from Examples adds to Helpful Classroom Sorter and keep the

> evidence.

Create the first small labeled example set for the sorter and explain which visual differences it covers.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Count each candidate from zero

> Count each candidate from zero. Apply the supplied Learning from Examples facts and keep your reasoning.

Heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Predict all four labels under threshold 3, then under threshold 7, using tall when height is at least the threshold.

List the disagreeing examples and separate totals.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Explain a tied fit

> Explain a tied fit. Apply the supplied Learning from Examples facts and keep your reasoning.

Thresholds 5 and 6 both have zero disagreements on the W1 **fitting set**. The **tie rule** chooses the smaller threshold. Which is selected? Give a whole-number height where the two candidates disagree and both **prediction**s.

Explain what the **tie rule** does not prove.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Update with an added example

> Update with an added example. Apply the supplied Learning from Examples facts and keep your reasoning.

Add height 5 cm labelled short to the four **fitting** examples. Recompute the disagreement totals for thresholds 3, 5, 6 and 7. Select the candidate with the fewest disagreements, keeping the smaller-threshold **tie rule**.

Explain why the new label is part of **fitting**.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Fit a new distance task

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit a new distance task. Apply the supplied Learning from Examples facts and keep your reasoning.

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My reasoning and evidence

W5 Keep conflicting records visible

> Keep conflicting records visible. Apply the supplied Learning from Examples facts and keep your reasoning.

Two **fitting** records both have height 6 cm. One is labelled short and the other tall. A learner says a clever height-only rule can agree with both on every run.

Explain the problem for a deterministic rule and name two checks to make before changing either record.

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Learning from Examples

> A fictional classifier learns whether measured sticks are short or long. Explain the labelled step before checking the changed condition.

A fictional classifier learns whether measured sticks are short or long.

Training lengths 2,4,8,10 cm have labels short,short,long,long.

Candidate boundaries: long at least 5 or at least 7. Lowest errors wins; tie chooses smaller boundary.

AI ACADEMY / AI-1/13

Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/13

Learning from Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training rows

2 and 4 are short; 8 and 10 are long.



2. Candidate checks

Boundaries 5 and 7 both predict S,S,L,L: zero errors each.



3. Tie policy

Choose boundary 5 because the declared tie rule prefers the smaller value.



4. New prediction

Length 6 is predicted long by the selected rule.

CHANGE ONE THING

Reverse the tie policy to prefer the larger boundary.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which new measured length would reveal disagreement between the candidates?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save candidate definitions, **training** errors and tie policy with any sorter model version.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Save candidate definitions, **training** errors and tie policy with any sorter model version.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classifier learns whether measured sticks are short or long.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Length 6 is predicted long by the selected rule.

1. **Training** rows

2. New **prediction**

3. Tie policy

4. Candidate checks

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: New **prediction** Length 6 is predicted long by the selected rule. There is no **training** stick between 4 and 8 to distinguish these candidates.

The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true. Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Reverse the tie policy to prefer the larger boundary. Which

conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Boundary 7 is selected and predicts short at 6.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Boundary 7 is selected and predicts short at 6. The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Repair a missing building block

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

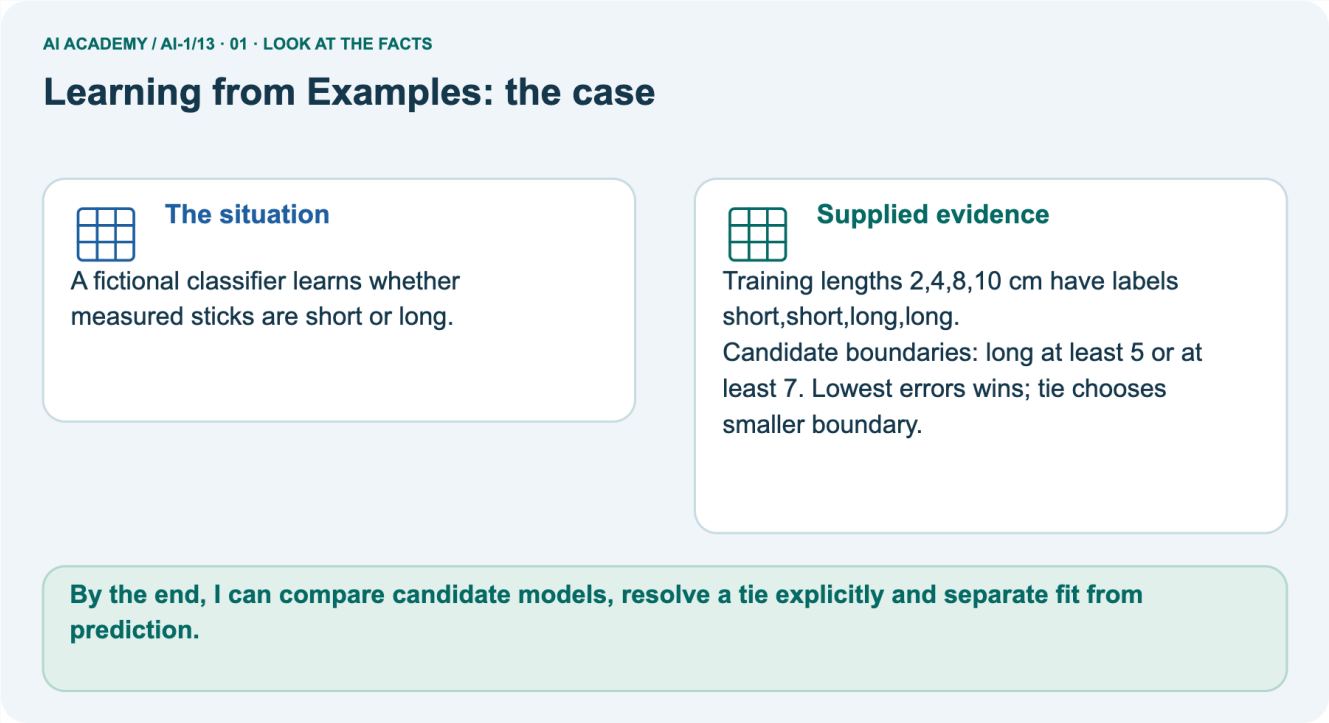
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Learning from Examples picture from its labels, then record a relevant limit.

01. Learning from Examples: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

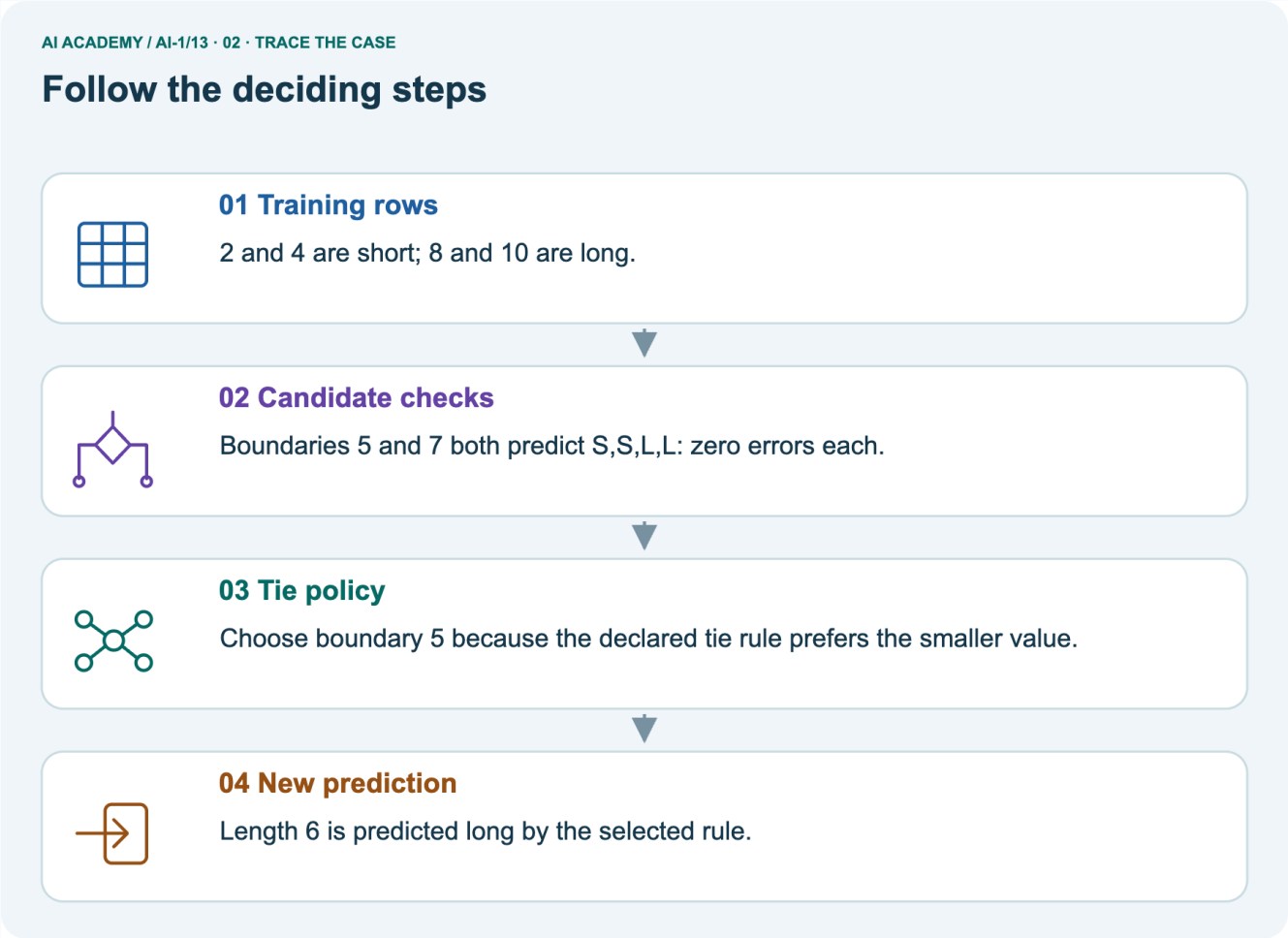
A fictional classifier learns whether measured sticks are short or long. Supplied facts: **Training** lengths 2,4,8,10 cm have labels short,short,long,long. Candidate boundaries: long at least 5 or at least 7.

Lowest errors wins; tie chooses smaller boundary.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



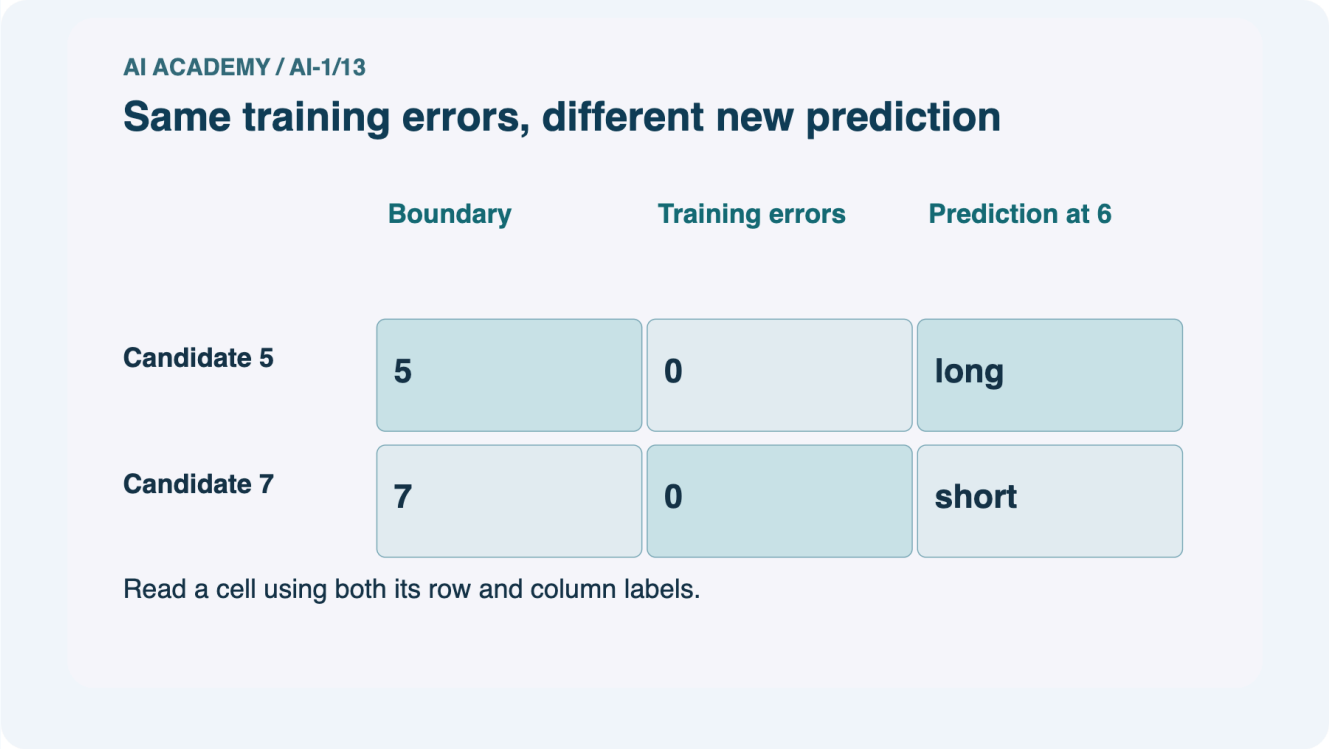
Follow the labelled picture with the matching explanation. Open the vector diagram

Training rows: 2 and 4 are short; 8 and 10 are long. → Candidate checks: Boundaries 5 and 7 both predict S,S,L,L: zero errors each. → Tie policy: Choose boundary 5 because the declared **tie rule** prefers the smaller value. → New **prediction**: Length 6 is predicted long by the selected rule.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same training errors, different new prediction



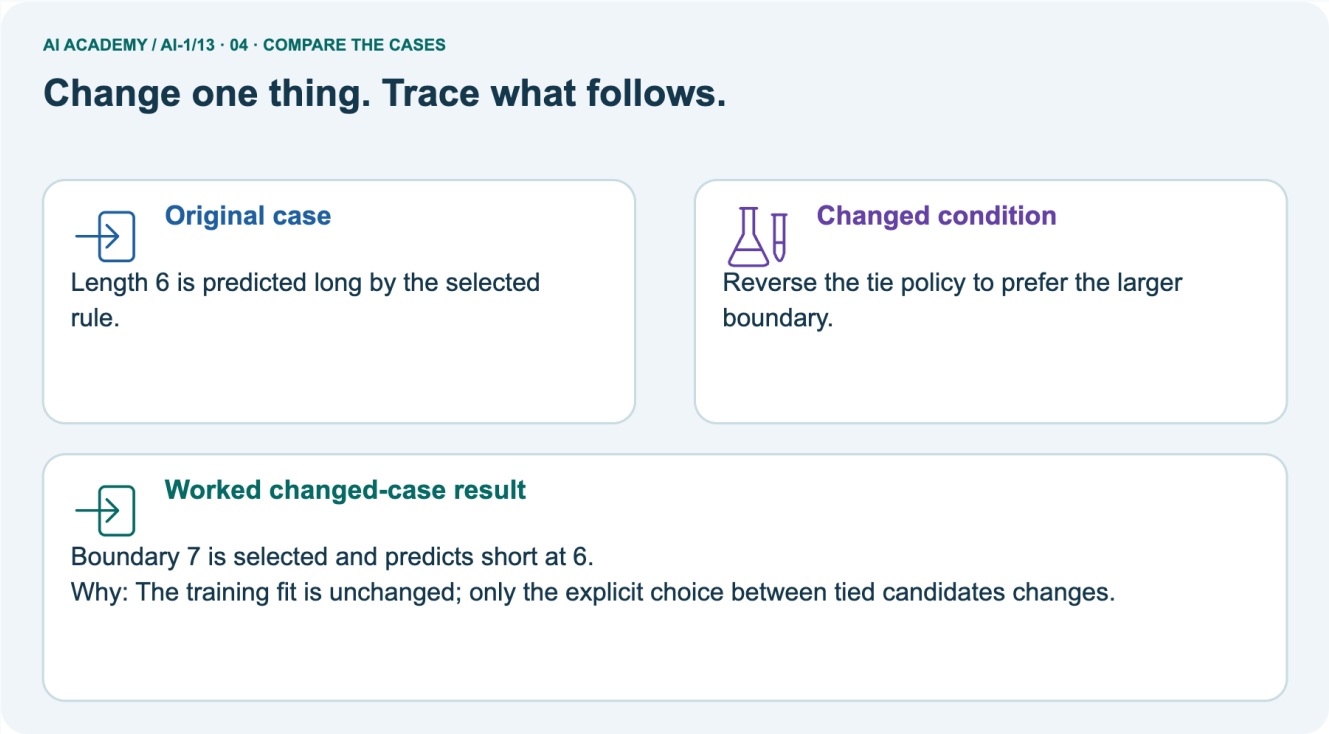
Follow the labelled picture with the matching explanation. Open the vector diagram

Same **training** errors, different new **prediction**. Candidate 5: Boundary = 5, **Training** errors = 0, **Prediction** at 6 = long. Candidate 7: Boundary = 7, **Training** errors = 0, **Prediction** at 6 = short.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

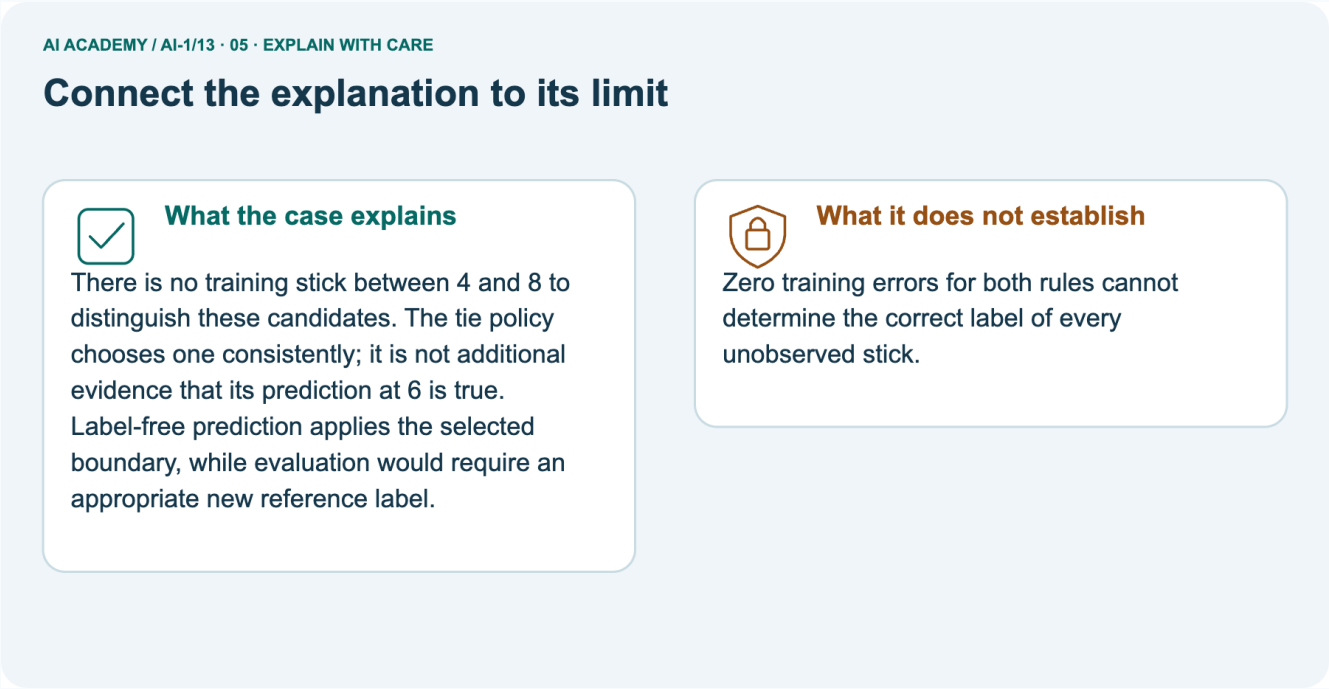
Original case: Length 6 is predicted long by the selected rule. Changed condition: Reverse the tie policy to prefer the larger boundary. Expected result: Boundary 7 is selected and predicts short at 6.

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

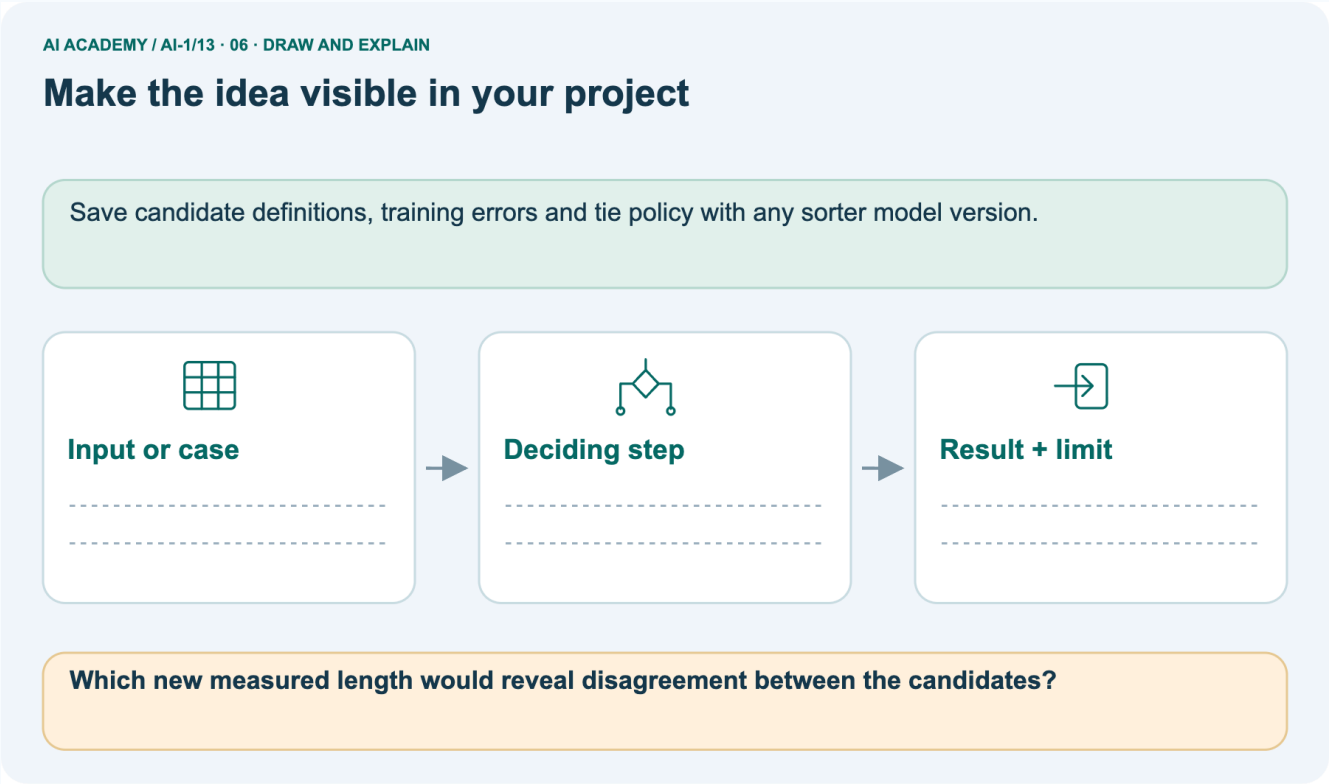
There is no **training** stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true.

Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label. Limit: Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save candidate definitions, **training** errors and tie policy with any sorter model version. Draw your **input**, deciding step, result and limit.

Supported question: Which new measured length would reveal disagreement between the candidates?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.